

Basic IO work

- 1) Write a program that asks the user to enter:
 - a. The first **name** of a friend
 - b. A favourite **activity**
 - c. A favourite **colour**
 - d. An **emotion**then prints the following line (you should replace the items in italics with the values entered):
I had a dream that *name* went *activity* and turned *colour* from *emotion*!
- 2) Write a program that calculates the number of **whole** tiles required to cover a wall of user-specified dimensions. The area of a single tile is 3*3 inches (9 inches). Your program should take in the height and width of the user's wall and output the number of tiles needed to cover this area, ignoring the need for grouting etc.

Branching/Decisions (If and switch statements)

- 3) Write a program that models a basic calculator. Your program should ask the user to enter two numbers, then prompt the user to choose from the following options:
 - Addition
 - Subtraction
 - Division
 - Multiplication

Your program should output the answer to the chosen type of calculation with the supplied numbers, i.e. if the user entered 3 and 5 as the initial numbers, then selected Multiplication, the answer displayed would be 15.

- 4) Write a second solution to question 3 using a switch statement to select the appropriate operation (addition, subtraction, division or multiplication)
 - a. *Note: If you wrote your solution to question 3 using a switch statement, use an if statement here!*

Repetition/Iteration (loops)

- 5) Write a program that takes in numbers from the user. Input should terminate when the user enters -1. The program should then print out the sum and average of the entered numbers (-1 should not be included in these calculations)
- 6) Write a program that takes in two numbers from the user. The program should then add up all the numbers between those the user entered (inclusive) and output the result. For example, if a user enters 3 and 7, then the program should print out 25 (i.e. 3+4+5+6+7).
Note: Your program should still work correctly, even if the user enters the bigger number first (i.e. 7 and then 3 will still calculate 3+4+5+6+7).
- 7) Update your calculator program to repeatedly run until the user enters "exit".

Arrays & Lists

- 8) Write a program that:
 - a. Asks a user how many items they wish to buy

- b. Creates an **array** of doubles with the specified number of slots
 - c. Creates an **array** of Strings with the specified number of slots
 - d. For each slot:
 - i. Takes in the name of the item being bought and stores it
 - ii. Takes in the price of the item being bought and stores it
 - e. Once all items have been entered, your program should loop through the stored data to print out a “receipt”. Each line should contain the name of the item and its price.
- 9) Update your shopping program to find the name of all items priced over €5. This should store the names in an **arraylist** as well as displaying them to the user.

Methods

- 10) Move the logic for finding the name of all items priced over X in your shopping program to a method called **findGreater()**. This method should take in two arrays (the array of names and the array of prices) and a double (the price threshold to be used) and should return the **arraylist of product names**.
- 11) Write a program to input 3 integers (separated by spaces) into 3 variables and output the maximum of the 3 values. Your solution should contain the following methods:
- a. **getValidInteger()** which takes in a *prompt* as a parameter and returns a valid integer.
 - b. **findMax()** which takes in 3 *integer* parameters and returns the biggest number.
 - c. **displayOutput()** which takes in a *string to be displayed* as a parameter. The method should display the supplied text on the console and should return nothing.
- 12) Create a class called **InputUtility** that contains the following methods:
- a. **getValidInteger()** which takes in a *prompt* as a parameter and returns a valid integer (this is the same as the method you wrote in exercise 11, just in a different class)
 - b. **getValidInteger()** which takes in THREE parameters:
 - i. A prompt to be displayed to the user
 - ii. An upper bound/limit on the allowable value
 - iii. A lower bound/limit on the allowable valueThe method should return a valid integer between the 2 specified values (inclusive)
- 13) Modify your answer to question 11 to use the **getValidInteger()** method from your InputUtility class.